NON POWER NUCLEAR APPLICATIONS

Research Reactors

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- Many of the world's nuclear reactors are used for research and training, materials testing, or the production of radioisotopes for medicine and industry. They are basically neutron factories.
- These are much smaller than power reactors or those propelling ships, and many are on university campuses.
- There are about 220 such reactors operating, in 53 countries.
- Some operate with high-enriched uranium fuel, and international efforts have substituted high-assay low-enriched fuel in most of these. Some radioisotope production also uses high-enriched uranium as target material for neutrons, and this is being phased out in favour of low-enriched uranium.

Research reactors comprise a wide range of civil and commercial nuclear reactors which are generally not used for power generation. The term is used here to include test reactors, which are more powerful than most. The primary purpose of research reactors is to provide a neutron source for research and other purposes. Their output (neutron beams) can have different characteristics depending on use. They are small relative to power reactors whose primary function is to produce heat to make useful amounts of electricity. They are essentially net energy users. Their power is designated in megawatts (or kilowatts) thermal (MWth or MWt), but here we will use simply MW (or kW). Most range up to 100

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MW, compared with 3000 MW (*i.e.* 1000 MWe) for a typical power reactor. In fact the total power of the world's 250 research reactors is little over 3000 MW.

Research reactors are simpler than power reactors and operate at lower temperatures. They need far less fuel, and far less fission products build up as the fuel is used. On the other hand, their fuel requires uranium that is more highly enriched, typically up to 20% U-235, known as high-assay low-enriched uranium (HALEU), although some older ones still use 93% U-235. They also have a very high power density in the core, which requires special design features. Like power reactors, the core needs cooling, usually passively and only the higher-powered test reactors need forced cooling. Usually a moderator is required to slow down the neutrons and enhance fission. As neutron production is their main function, most research reactors also need a reflector to reduce neutron loss from the core.

Operational research reactors

	Operational research reactors
Russia	52
USA	50
China	16
India	7
Argentina	5
Canada	5
Germany	5
Italy	5
Brazil	4
Iran	4
Kazakhstan	4
Belarus	3
Belgium	3
Czech Republic	3
France	3
Indonesia	3

	Operational research reactors
Japan	3
Ukraine	3
Others	45
Total	223

Nearly all of the world's research reactors operate with thermal (slow) neutrons; Russia claims that its BOR-60 at Dimitrovgrad is the only fast neutron research reactor.* It started up in 1969 and is to be replaced after the end of 2020 with the MBIR, with four times the irradiation capacity. There is a world shortage of fast reactor research capacity, especially for fast neutron materials testing for Generation IV reactor developments. In February 2018 a bipartisan bill passed by the US House of Representatives authorized $2 billion for the construction of a "versatile reactor-based fast neutron source, which shall operate as a national user facility" by 2026. It will be a research reactor for "development of advance reactor designs, materials and nuclear fuels" of at least 300 MWt.

* Several small experimental reactors – CEFR, FBTR, Joyo – fall into the broad category of research reactors in that they are not designed to produce power for the grid, but they do not generally operate as neutron irradiation and research facilities for third parties (although CEFR may do so to some extent).

As of June 2021 the IAEA research reactor database showed that there were 223 operational research reactors, 11 under construction, 16 planned, 27 temporarily or in extended shutdown, 58 permanently shutdown, and 511 decommissioned or undergoing decommissioning. A majority of the operational and temporary shutdown research reactors are over 40 years old.

Many research reactors are used with international collaboration, and the products of those used for isotope production are traded internationally. The IAEA has designated two international research hubs based on research reactors, giving them ICERR (International Centres based on Research Reactors) status valid for five years. The first is in France, based on CEA's Saclay and Cadrache facilities, the second is in Russia, the Research Institute of Atomic Reactors (RIAR) at Dimitrovgrad, with six research reactors available to IAEA member states.

Types of research reactor

There is a much wider array of designs in use for research reactors than for power reactors, where 80% of the world's plants are of just two similar types. They also have different operating modes, producing energy which may be steady or pulsed.

A common design (47 units) is the pool-type reactor, where the core is a cluster of fuel elements sitting in a large pool of water. Among the fuel elements are control rods and empty channels for experimental materials. Each element comprises several (e.g. 18) curved aluminium-clad fuel plates in a vertical box. The water both moderates and cools the reactor, and graphite or beryllium is generally used for the reflector, although other materials may also be used. Apertures to access the neutron beams are set in the wall of the pool. Tank type research reactors (21 units) are similar, except that cooling is more active.

The TRIGA (Training, Research, Isotopes, General Atomics) reactor is a common pool-type design (36 units) with three generations of design commissioned since 1960. The core consists of 60-100 cylindrical fuel elements about 37 mm in diameter, 722 mm long with aluminium or steel cladding enclosing self-moderating uranium zirconium hydride fuel. Enrichment today is HALEU - under 20%. The core sits in a pool of water and generally uses graphite as a reflector. This kind of reactor can safely be pulsed to very high power levels (up to 22,000 MW) for fractions of a second. Its UZrH fuel has a very strong prompt negative temperature coefficient, and a rapid increase in power is quickly cut short by the negative reactivity effect of the fuel.

Other designs are moderated by heavy water (c. 10 units) or graphite. A few are fast reactors, which require no moderator and can use a mixture of uranium and plutonium as fuel. Homogenous type reactors have a core comprising a solution of uranium salts as a liquid, contained in a tank about 300 mm diameter. The simple design made them popular early on, but only a few are now operating, and all except two of those (20 kW) are very low power.

Research reactors have a wide range of uses, including analysis and testing of materials, and production of radioisotopes. Their capabilities are applied in many fields, within the nuclear industry as well as in fusion research, environmental science, advanced materials development, drug design and nuclear medicine.

Four high-flux research reactors are in operation: the 45 MW High Flux Reactor in Petten (Netherlands), which is nearing the end of its design life, the upgraded 85 MW High Flux Isotope Reactor at Oak Ridge National Laboratory (USA), the 20 MW Heinz Maier-Leibnitz FRM II (Germany) and the 100 MW PIK high-flux reactor at St Petersburg (Russia).

Most research reactors (over 160) are principally for research, although some may also produce radioisotopes. As expensive scientific facilities, they tend to be multi-purpose, and many have been operating for more than 30 years.

About 820 research and test reactors have been built worldwide, 307 of these in the USA and 121 in Russia.

In 2020, Russia has most operational research reactors, followed by the USA (see above table). Many research reactors were built in the 1960s and 1970s. The peak number operating was in 1975, with 373 in 55 countries.

Uses

Neutron beams are uniquely suited to studying the structure and dynamics of materials at the atomic level. Neutron scattering is used to examine samples under different conditions such as variations in vacuum pressure, high temperature, low temperature and magnetic field, essentially under real-world conditions.

Using neutron activation analysis, it is possible to measure minute quantities of an element. Atoms in a sample are made radioactive by exposure to neutrons in a reactor. The characteristic radiation each element emits can then be detected.

Neutron activation is the only common way that a stable material can be made radioactive. It is used to produce the radioisotopes, widely used in industry and medicine, by bombarding particular elements with neutrons so that the target nucleus has a neutron added. For example, yttrium-90 microspheres to treat liver cancer are produced by bombarding yttrium-89 with neutrons.

Neutron activation can result in fission. The most widely used isotope in nuclear medicine is technetium-99m, a decay product of molybdenum-99*. This is usually produced by irradiating a target of uranium foil with neutrons (for a week or so) and then separating the molybdenum-99 from the other fission products of U-235 in a hot cell – the Mo-99 being about 6% of the fission products. Most Mo-99/Tc-99m production has been using HEU targets, but increasingly LEU is favoured and HEU is being phased out. (Mo-99 may also be produced in power reactors by neutron activation/capture in Mo-98 targets.)

* Technetium generators, a lead pot enclosing a glass tube containing the radioisotope, are supplied to hospitals from the nuclear reactor where the isotopes are made. They contain molybdenum-99, with a half-life of 66 hours, which progressively decays to technetium-99m, with a half-life of 6 hours. The Tc-99 is washed out of the lead pot by saline solution when it is required. It is then attached to a particular protein for administering to the patient. After two weeks or less the generator is returned for recharging, since it loses 22% of its product every 24 hours.

Research reactors can also be used for industrial processing. **Neutron transmutation doping** (NTD) changes the properties of silicon, making it highly conductive of electricity. Large, single crystals of silicon shaped into ingots, are irradiated inside a reactor reflector vessel. Here the neutrons change one atom of silicon in every billion to phosphorus. The irradiated silicon is sliced into chips and used for a wide variety of advanced computer applications. NTD increases the efficiency of the silicon in conducting electricity, an essential characteristic for the electronics industry.

In **materials testing reactors** (MTRs), materials are also subject to intense neutron irradiation to study changes. For instance, some steels become brittle, and alloys which resist embrittlement must be used in nuclear reactors. The international project to build the 100 MWt Jules Horowitz reactor at Cadrache in France will enable research on materials which will be vital in Generation IV nuclear plants. It is designed to produce very high neutron flux – about twice that of France's Osiris MTR. Civil works were more than 80% complete at the end of 2014 and test loops are being fitted out. It is due to be operational in 2021. At present Belgium's BR2 is the largest MTR in Europe.

A very small US reactor is designed to perform research and development on various operational features of microreactors to improve their integration with end-user applications. The Department of Energy plans to build the Microreactor Applications Research Validation and Evaluation (MARVEL) reactor of 100 kWt

at Idaho National Laboratory's Transient Reactor Test Facility (TREAT). It will be based on the 1965 SNAP-10A – the only US fission reactor to have been launched into space. Fuel will be TRIGA HALEU UZrH, cooling will be by sodium or salt for operating temperature of 500-550°C. Power conversion will be by Stirling engines. This will be the first US small-scale reactor for such R&D purposes in 40 years. The DOE Microreactor Program is focused on very small, factory fabricated, transportable reactors to provide power and heat for decentralized generation in civilian, industrial and defence energy sectors.

Like power reactors, research reactors are covered by IAEA safety inspections and safeguards, because of their potential for making nuclear weapons. India's 1974 explosion was the result of plutonium production in a large, but internationally unsupervised, research reactor which closed at the end of 2010.

One of the more interesting and powerful test reactors was Plum Brook in Ohio, USA, which operated for NASA over 1961-73 and was designed to research nuclear power for aircraft, then nuclear-powered rockets and spacecraft. It was 60 MW pool-type, light water cooled and moderated, with a very high neutron flux – 420 trillion/cm^2/s.

See also paper on Australian Research Reactors.

Fuel

Research reactor fuel is more highly enriched (typically about 20% today) than power reactor fuel. It is high-assay low-enriched uranium (HALEU), which means it has less U-238, hence the used fuel has fewer actinides and less heat from radioactive decay. The proportion of fission products is not much different from used power reactor fuel.

Fuel assemblies are typically plates or cylinders of uranium-aluminium alloy (U-Al) clad with pure aluminium. They are different from the ceramic UO_2 pellets enclosed in Zircaloy cladding used in power reactors. Only a few kilograms of uranium is needed to fuel a research reactor, albeit more highly enriched, compared with perhaps a hundred tonnes in a power reactor. Research reactors typically operate at low temperatures (coolant below 100°C), but the operating conditions are severe in other ways. While power reactor fuel operates at power density of about 5 kW/cc, a research reactor fuel may be at 17 kW/cc in the fuel meat. Also burnup is

very much higher, so the fuel must withstand structural damage from fission and accommodate more fission products.

Five Russian-designed research reactors in Russia use high-enriched uranyl sulfate liquid fuel. One in Uzbekistan was decommissioned in 2014 and the fuel airlifted to Mayak.

Highly-enriched uranium (HEU – >20% U-235) allowed more compact cores, with high neutron fluxes and also longer times between refuelling. Therefore many reactors up to the late 1970s used it, and most state-of-the-art reactors had 93% enriched fuel. A total of 171 were built with a HEU core, and by 2020, 71 of these had been converted to use LEU and 28 had been shut down. The IAEA said "work remains to be done" on 72 still using HEU.

Since the early 1970s security concerns have grown, especially since many research reactors are located at universities and other civilian locations with much lower security than military weapons establishments where much larger quantities of HEU exist. Since 1978 only one reactor, the FRM-II at Garching in Germany, has been built with HEU fuel, while more than 20 have been commissioned on LEU fuel in 16 countries. The Jules Horowitz reactor in France will start up on uranium silicide fuel enriched to 27%, since the planned high-density U-Mo fuel will not be ready in time for it.)

The question of enrichment was a major focus of the UN-sponsored International Nuclear Fuel Cycle Evaluation in 1980. It concluded that to guard against weapons proliferation from the HEU fuels then commonly used in research reactors, enrichment should be reduced to no more than 20% U-235, *i.e.* HALEU. This followed a similar initiative by the USA in 1978 when its programme for **Reduced Enrichment for Research and Test Reactors** (RERTR) was launched.

Most research reactors using HEU fuel were supplied by the USA and Russia, hence efforts to deal with the problem are largely their initiative. The RERTR programme concentrates on reactors over 1 MW which have significant fuel requirements. However, some are defence-related (mostly in Russia) or impractical for other reasons. Some have lifetime cores which require no refuelling, so there is little incentive to convert them. In February 2020 the IAEA announced that almost 3500 kg of HEU fuel had been removed from research reactor sites around the world.

In 2004 the US National Nuclear Security Administration (NNSA) set up the Global Threat Reduction Initiative (GTRI), which is congruent with RERTR objectives though it is mainly tackling the

disposition of HEU fuel (fresh and used) and other radiological materials. RERTR is now a major part of GTRI. GTRI claims accelerated removal of Russian-origin fresh and used HEU fuel to Russia and of US-origin fuel to the USA.

In particular, to January 2010, 915 kg of fresh and used HEU fuel had been returned to Russia from at least 11 countries including Hungary (155 kg), Serbia, Romania, Libya, Uzbekistan, Poland, Czech Republic, Latvia and Vietnam. In 2016, the last of 700 kg of HEU fuel from Poland joined them after Poland's 30 MWt Maria reactor was fully converted to HALEU in 2014. And to January 2010, 1240 kg of US-origin HEU fuel had been returned from Europe, Israel, Turkey, Latin America, Japan and Southeast Asia. Since then more has come from some of these countries and from Belgium, Italy, Chile, Mexico, Ukraine, South Africa and Austria. In mid-2016 several hundred kilograms more was returned from Japan's Fast Critical Assembly, along with some plutonium. The HEU returned to the USA is downblended, with the plutonium disposed of at the Waste Isolation Pilot Plant (WIPP) in New Mexico. In 2011, 33kg of HEU in Kazakhstan was downblended to LEU there and returned to the Institute of Nuclear Physics for use once the WWR-K reactor was converted to use it. Over 2015-19 more than 200 kg of HEU from two research reactors at Canada's Chalk River, Ontario, was moved to Savannah River, South Carolina.

After a hiatus of six years the US government late in 2008 had converted five university research reactors from using high- to low-enriched uranium fuel.* It was reported in 2006 that worldwide, 40 remained to be converted under the RERTR scheme using currently-available fuels, and 19 more await development of high-density fuel. At the end of 2017 these included the Advanced Test Reactor and the Advanced Test Reactor Critical Assembly, both at Idaho National Laboratory; the High Flux Isotope Reactor at Oak Ridge National Laboratory, Tennessee; the Massachusetts Institute of Technology Reactor (MITR) in Cambridge, Massachusetts; the National Bureau of Standards Reactor in Gaithersburg, Maryland; and the University of Missouri Research Reactor (MURR) in Columbia, Missouri.

* Texas A&M, University of Florida, Purdue, Oregon State and Washington State University reactors can now operate on fuel of less than 20% enrichment, and the University of Wisconsin reactor was to be converted in 2009.

China converted the first of its miniature neutron source reactors (MNSRs) to a denser LEU fuel in 2016, in partnership with the US

NNSA and Argonne National Laboratory. Other MNSRs are in China, Ghana, Nigeria, Iran, Pakistan and Syria. The MNSR core has about one kilogram of of 90% enriched fuel. HEU from Ghana and Nigeria at least has been returned to the China Institute of Atomic Energy.

By December 2018 NNSA's Material Management and Minimization (M3) Reactor Conversion Program had converted or verified shutdown of more than 90 HEU-fuelled research and test reactors worldwide over the past 30 years, and had confirmed the disposition of more than 6.7 tonnes of HEU and plutonium from 33 countries.

New LEU fuels

These RERTR programmes have led to the development and qualification of new, high density, high-assay low enriched uranium (HALEU) fuels. The original fuel density was about 1.3-1.7 g/cm^3 uranium. Lowering the enrichment meant that the density had to be increased. Initially this was to 2.3-3.2 g/cm^3 with existing U-Al fuel types.

To September 2009, 67 research reactors (17 in USA) had been converted to low-enriched uranium silicide fuel or shut down, including major reactors in Ukraine, Uzbekistan and South Africa. Another 34 are convertible using present fuels. A further 28, mostly Russian designs but including two US university reactors, need higher-density fuels not yet available. The goal was to convert or shut 129 reactors by 2018. US exports of HEU declined from 700 kg/yr in mid 1970s to almost zero by 1993.

The Soviet Union made similar efforts from 1978, and produced fuel of 2.5 g/cm3 with enrichment reduced from 90 to 36%. It largely stopped exports of 90% enriched fuel in the 1980s. No Russian research reactor has yet been converted to LEU, and the Russian effort has been focused on its reactors in other countries. However, Russia has been looking at the feasibility of converting six domestic reactors,* while others will require high-density fuels. Early in 2012 a joint Rosatom-US NNSA project completed studies on converting two of these six reactors to HALEU fuel. Another 68 Russian reactors fall outside the scope of the conversion program because they are defence-related or special purpose. A December 2010 agreement with the US DOE relates to feasibility studies on converting six Russian research reactors to HALEU.

* IR-8, OR, and Argus at Kurchatov Institute, IRT-MEPHI at Moscow Engineering Physics Inst, IRT-T at Tomsk Polytechnic Inst, MIR at Dimitrovgrad Research Inst.

The first generation of new HALEU fuels used uranium and silicon (U_3Si_2-Al – uranium silicide dispersed in aluminium) – at 4.8 g/cm^3. There have been successful tests with denser U_3Si-Al fuel plates up to 6.1 g/cm^3, but US development of these silicide fuels ceased in 1989 and did not recommence until 1996. In Russia the Bochvar Institute is developing fabrication technology for uranium disilicide pellets as part of developing accident-tolerant fuels for power reactors. As well as high uranium density, it points to high thermal conductivity and low heat capacity of silicide fuel. The presence of silicon makes reprocessing more difficult.

An international effort is underway to develop, qualify and license a high-density fuel based on U-Mo alloy dispersed in aluminium, with a density of 6-8g/cm^3. The principal organizations involved are the US RERTR program at the Idaho National laboratory (INL) since 1996, the French U-Mo Group (CEA, CERCA, COGEMA, Framatome-ANP and Technicatome) since 1999 and the Argentine Atomic Energy Commission (CNEA) since 2000. This development work was undertaken to provide fuels which can extend the use of LEU to those reactors requiring higher densities than available in silicide dispersions and to provide a fuel that can be more easily reprocessed than the silicide type. Approval of this fuel was expected in 2006 but tests since 2003 failed to confirm performance due to unstable swelling under high irradiation. Attention has turned to monolithic U-Mo fuel.

In 2012, Belgium, France, South Korea and the USA agreed to cooperate in the development of high-density HALEU fuel production technology using centrifugal atomization technology developed by the Korea Atomic Energy Research Institute (KAERI). The USA provided 110 kg of HALEU in June 2013 for KAERI to manufacture 100 kg of atomized U-Mo powder. In January 2014, the powder was shipped to France for fabrication into fuel elements by Areva's research reactor fuel manufacturer CERCA. Testing of this monolithiic HALEU-Mo fuel began in the Advanced Test Reactor at Idaho National Laboratory (INL) in October 2015, and in April 2017 KAERI announced the successful completion of those verification tests.

This monolithic HALEU-Mo fuel is essentially pure metal with 10% Mo, instead of a dispersion of U-Mo in aluminium. The uranium density is 15.6 g/cm^3 and this would enable every research reactor in the world to convert from HEU to HALEU fuel without loss of

performance. In December 2017 the NNSA presented a preliminary report of HALEU-Mo monolithic fuel to the Nuclear Regulatory Commission as a first step in qualifying it, and being able to convert the six remaining US high-performance research reactors from HEU fuel. In February 2020 NNSA awarded a contract to BWX Technologies to prepare for HALEU-Mo fuel production at Lynchburg, Virginia.

KAERI said the results from the testing of the fuel will be used to obtain a construction licence for a new research reactor planned in Busan, which it hopes will be the first application for this HALEU-Mo alloy fuel.

In Russia, a parallel Russian RERTR programme funded jointly by Rosatom and the US RERTR programme has been working since 1999 to develop U-Mo dispersion fuel with a density of 2-6 g/cm^3 for use in Russian-designed research and test reactors. However, this has not fulfilled expectations.

All solid fuel is aluminium-clad.

Used fuel

Used fuel from research reactors usually generates less than 2kW/m^3 of decay heat, so is classed as intermediate-level waste (ILW), though the activity may still be quite high.

U-Al fuels can be reprocessed by Areva in France, and U-Mo fuels may also be reprocessed there. U-Si and TRIGA fuels are not readily reprocessed in conventional facilities. However, at least one commercial operator has confirmed that U-Si fuels may be reprocessed in existing plants if diluted with appropriate quantities of other fuels, such as U-Al.

To answer concerns about interim storage of spent research fuel around the world, the USA launched a program to take back US-origin spent fuel for disposal and nearly half a tonne of U-235 from such HEU fuel has been returned. By the time the program was to end with fuel discharged in 2006, U-Mo fuel was expected to be available. Due to the slippage in target date, the US take-back program has now been extended by ten years.

Disposal of high-enriched or even 20% enriched fuel needs to address problems of criticality and requires the use of neutron absorbers or diluting or spreading it out in some way.

In Russia, a parallel trilateral program involving IAEA and the USA to move 2 tonnes of HEU and 2.5 tonnes of LEU spent fuel to the Mayak reprocessing complex near Chelyabinsk over the ten years to 2012. This Russian Research Reactor Fuel Return Program (RRR FRT) envisaged 38 shipments (of both fresh and used fuel) from ten countries over 2005-08, then 8+ shipments from six countries to remove all HEU fuel discharged before reactors converted to HALEU or shut down. Seventeen countries have Soviet-supplied research reactors, and there are 25 such reactors outside Russia, 15 of them still operational. Since Libya joined the program in 2004, only North Korea objects to it.

The 2004 joint US-Russian program to retrieve used fuel from 14 countries (Belarus, Bulgaria, Hungary, Vietnam, Germany, Kazakhstan, Latvia, Libya, Poland, Romania, Serbia, Uzbekistan, Ukraine and the Czech Republic) has been extended from 2016 to 2024.

Aqueous homogeneous reactors

Aqueous homogeneous reactors (AHRs) have the fuel mixed with the moderator as a liquid. Typically, low-enriched uranium nitrate is in aqueous solution. About 30 AHRs have been built as research reactors and have the advantage of being self-regulating and having the fission products continuously removed from the circulating fuel. However, corrosion problems and the propensity of water to decompose radiolytically (due to fission fragments) releasing gas bubbles have been design problems.

A series of three reactors were built at Los Alamos National Laboratory in the mid-19402/early 1950s. The first AHR at Oak Ridge National Laboratory went critical in 1952, and attained full power of one megawatt in 1953. A second one there reached 5 MW in 1958. Plans for a 70 to 150 MWe commercial unit did not proceed. A 1 MWt AHR operated in the Netherlands 1974-77 using Th-HEU MOX fuel. At Russia's Kurchatov Institute the 20 kW ARGUS AHR has operated since 1981, and R&D on producing Sr-89 and Mo-99 from it is ongoing. The Mo-99 is extremely pure, making the design potentially valuable for its commercial production. As of 2006, only five AHRs were operating, but the concept of extracting medical isotopes directly from the fuel has sparked renewed interest in them. The USA, China and Russia are assessing the prospects of using AHRs for commercial radioisotope production.

In 2008, the IAEA summarized: "The use of solution reactors for the production of medical isotopes is potentially advantageous because of their low cost, small critical mass, inherent passive safety, and simplified fuel handling, processing and purification characteristics. These advantages stem partly from the fluid nature of the fuel and partly from the homogeneous mixture of the fuel and moderator in that an aqueous homogeneous reactor combines the attributes of liquid fuel homogeneous reactors with those of water moderated heterogeneous reactors. If practical methods for handling a radioactive aqueous fuel system are implemented, the inherent simplicity of this type of reactor should result in considerable economic gains in the production of medical isotopes." Thermal power can be 50-300 MW at low temperature and pressure, and low enriched uranium fuel used. However, recovering desired isotopes on a continuous production basis remains to be demonstrated. As well as those in solution, a number of volatile radioisotopes used in nuclear medicine can be recovered from the off-gas arising from radiolytic 'boiling'. For isotopes such a Sr-89 this is very much more efficient than alternative production methods.

At the end of 2007, Babcock & Wilcox (B&W) notified the US Nuclear Regulatory Commission that it intended to apply for a licence to construct and operate a Medical Isotope Production System (MIPS) – an AHR system with low-enriched uranium in small 100-200 kW units for Mo-99 production. A single production facility could have four such reactors. B&W expects a five-year lead time to first production. The fuel is brought to criticality in a 200-litre vessel. As fission proceeds, the solution is circulated through an extraction facility to remove the Mo-99 and then back into the reactor vessel, which is at low temperature and pressure. In January 2009, B&W Technical Services Group signed an agreement with radiopharmaceutical and medical device supplier Covidien to develop technology for the MIPS. In October 2012 Covidien pulled out of the joint venture with B&W “after learning that the time and cost involved with the project would be greater than originally expected.” B&W appears to have dropped the MIPS.

Research reactors originally with high-enriched uranium (HEU) fuel

	Type	Power - kW	Enrichment %	Source of fuel
Argentina	pool	500	90, now LEU	USA
Austria	Triga	250	20-70	USA

	Type	Power - kW	Enrichment %	Source of fuel
	Argonaut	10	20-90	USA
Belgium	tank	100,000	74-93	USA
Canada	pool	5,000	93	USA
	Slowpoke	20 (x 3)	93	USA
Chile	pool	2,000	90	France
	pool	5,000	20-45	USA
China	Crit fast	0.05	90	China
	tank	125,000	90	China
	MNSR	27	90	China
	pool	5000	90	China
	MNSR	30-33 (x 3)	90	China
Czech Rep	tank	10,000	36, now LEU	Russia
	pool	5	36	Russia
France	pool	0.1	90-93	USA
	Tank in pool	0.1	3-93	USA, France
	Crit fast	3	12-25	USA
	heavy water	58,300	93	USA
	pool	14,000	93	USA
	FBR – Phenix	563,000	22-28, now closed	France
	Argonaut	100	93	USA
	homogeneous	1	93	USA
Germany	pool	4,000	45-93	USA
	heavy water	23,000	80-93	USA
	pool	10,000	20-93	USA
	tank	0.01	36	Russia
Ghana	MNSR	30	90	China
Greece	pool	5	20-93	USA
Hungary	tank	10,000	36, now LEU	Russia

	Type	Power - kW	Enrichment %	Source of fuel
Israel	pool	5,000	93	USA
India	pool	1000	93	UK & France
	FBR	40,000	55-70	India
Iran	MNSR	30	90	China
Italy	Fast source	5	93	USA
Jamaica	Slowpoke	20	93	USA
Japan	Argonaut	0.01	90	USA
	tank	5000	93	USA
	Crit fast	2	20-93	USA, UK
	Tank	50,000	20-46	USA
	Crit assembly	0.1	45-93	USA
Korea, North	pool	8,000	36	Russia
Kazakhstan	pool	6,000	was 36, now LEU	Russia
	tank	10,000	90	Russia
	tank	60,000	90	Russia
Libya	pool	10,000	80, now LEU	Russia
Mexico	Triga	1000	20-70	USA
Netherlands	Argonaut	30	90	USA
	pool	2000	20-93	USA
Nigeria	MNSR	30	90	China
Pakistan	MNSR	30	90	China
Poland	pool	30,000	36-80	Russia
Portugal	pool	1000	93	USA
Romania	Triga	14,000	20-93	USA
Russia	various	(39 units, 12 being over 1 MW)	Various	Russia
South Africa	Tank in pool	20,000	87-93, now LEU	S.Africa
Sweden	pool	1000	93	USA
Switzerland	homogenous	2	90	USA

	Type	Power - kW	Enrichment %	Source of fuel
Syria	MNSR	30	90	China
UK	Fast burst	0.5	37.5	UK
	Pool	100	80	UK
Ukraine	tank	10,000	36, now LEU	Russia
USA	various	(22 units, 13 being 1 MW or more)	Various	USA
Uzbekistan	tank	10,000	36, now LEU	Russia
Vietnam	pool	500	36, now LEU	Russia
Serbia	heavy water	0.001	Up to 80	Russia
Total 39 countries		c 130 units		
Taiwan	pool	30	93	USA

Data from IAEA: Nuclear Research Reactors in the World, 2000, plus Nigeria.
NB some now are converted to LEU or closed.
MNSR = miniature neutron source reactor, Chinese copy of Slowpoke.
crit fast = very low power critical assembly for fast neutrons.

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